

MAHENDRA S H

Backend & LLMops Engineer | Python | Advanced SQL | Java | AWS

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Professional Summary

Data-driven Backend and LLMops Engineer with enterprise experience orchestrating high-throughput data pipelines, microservices, and AI telemetry systems. Specializes in LLM observability, root cause analysis (RCA), and tracking evaluation metrics (F1, data drift) for generative models. Proven ability to leverage **Advanced SQL**, **Python**, and **Power BI** to build operational dashboards, detect data anomalies, and optimize AWS cloud cost efficiencies for platforms processing **50k+ daily transactions**.

Technical Skills

LLMOps & Analytics:	Langfuse, Prompt Evaluation, Token Economics, F1/Accuracy Metrics, Model Drift Tracking, Power BI, Data Telemetry
Data Eng. & Cloud:	Advanced SQL, AWS (S3, SQS, SNS, Cost Explorer), ETL/ELT Pipelines, Data Modeling, Docker, CI/CD
Languages:	Python, Advanced SQL, Java 21, TypeScript, JavaScript
Backend & APIs:	Spring Boot, Spring Batch, Node.js, Express.js, RESTful APIs, Camunda BPM, OpenAPI
Databases:	PostgreSQL, MySQL, Prisma ORM, Query Optimization & Indexing

Work Experience

Infosys — Specialist Programmer / Software Developer | Jul 2024 – Jun 2026 | Hyderabad, India

- Engineered high-throughput enterprise data pipelines using **Java 21** and **Spring Batch**, automating the extraction, transformation, and verification of **50k+ daily transactional records**.
- Utilized **Advanced SQL** and **Python** to detect data trends, process anomalies, and identify backend performance bottlenecks, reducing API response times by **20%**.
- Analyzed **AWS SNS, SQS**, and **S3** telemetry and cost data to identify infrastructure optimization opportunities, managing cloud expenditures while maintaining massive system throughput.
- Performed root cause analysis (**RCA**) on system anomalies and tracked data evaluation metrics across CI/CD deployments and backend integrations.
- Built analytical **Power BI** dashboards for tracking accuracy, system usage, and cost analytics to improve operational decision-making for enterprise stakeholders.
- Integrated AWS services with **Camunda BPM** for asynchronous queue orchestration, reducing manual data verification overhead by **35%**.

Independent Software Development — Freelance Backend & Data Engineer | Jun 2023 – Jun 2024 | Mysore, India

- Engineered scalable backend architectures and SQL databases (**PostgreSQL, MySQL**) for bespoke enterprise platforms and client solutions.
- Integrated **Power BI** analytics into client administrative panels to visualize data trends, usage telemetry, and core business metrics.
- Developed secure **REST APIs** utilizing **Node.js** and **Express** to facilitate reliable data delivery and ingestion across multiple client environments.

Merav Infotech — Software Engineering Intern | Feb 2024 – May 2024 | Mysore, India

- Developed **RESTful APIs** and optimized data querying, reducing initial system payload load times by **15%**.
- Participated in **Agile** sprint planning, daily standups, code reviews, and cross-functional backend software delivery.

Selected Projects

ProjectX Generative AI Studio (LLMOps Pipeline) — Python, Next.js, Langfuse, AWS S3, SQL

- Implemented LLM observability pipelines using **Langfuse** and **Python** to continuously monitor model performance, token usage economics, and latency trends in production. Analyzed LLM extraction outputs against user-validated datasets, tracking evaluation metrics including F1 scores, accuracy, and data drift.

Cargo Readiness — CMA CGM Project — Java 21, Spring Batch, AWS SQS/SNS, PostgreSQL, Power BI

- Built event-driven backend workflows and ETL pipelines for container readiness assessment, successfully automating **50k+** daily booking data checks. Developed interactive **Power BI** operational dashboards fed by **Advanced SQL** queries to monitor system usage, process accuracy, and pipeline data anomalies.

SaaS Form Builder Data Platform — Node.js, PostgreSQL, AWS, TypeScript

- Architected a scalable backend data ingestion engine with secure **REST APIs** and analytics tracking, securely processing user inputs across **20+** embedded client sites.

SpendOptima Analytics Dashboard — TypeScript, PostgreSQL, SQL Analytics

- Built a comprehensive financial analytics dashboard featuring complex **SQL aggregations** and interactive timeseries visualizations for budget and expense telemetry.

Education

National Institute of Engineering (NIE) — Bachelor of Technology (B.Tech) in Computer Science & Engineering
CGPA: 7.3 / 10 | Relevant Coursework: Database Systems, Software Engineering, DBMS, Object-Oriented Design

Graduated May 2024
Mysore, India